

Mohd. Tabrez Aslam Mukadam

Mumbai, India · tabrezmukadam57@gmail.com · [LinkedIn](#) · tabrez.tech · [GitHub](#)

PROFILE

Cybersecurity researcher and BE undergraduate (CSE [IoT, Cyber Security & Blockchain]) specializing in **offensive security**, **smart contract auditing**, and **AI threat research**. Author of a peer-reviewed conference paper accepted at ICSSSD 2026. Disclosed 30+ vulnerabilities across 15+ platforms, holds a top 10% global CTF ranking, and is currently conducting AI/ML security research at **SwiftSafe**. Published technical writer across InfoSec Writeups, CoinsBench, and Block Magnates.

PUBLICATION

When Smart Systems Optimize the Wrong Thing: Evaluating Security and Sustainability in a Tourist Incident Response System

ICSSSD 2026 (Paper ID: 197) · Track: IoT & Sustainable Smart Systems · M.H. Saboo Siddik College of Engineering, Mumbai

- Designed a 3-layer adaptive IoT security framework (Technical, Socio-Technical, Temporal) for tourist incident response, simulating 500 trajectories across 25 urban zones and 2,143 synthetic panic alerts.
- Adaptive geofencing engine (Shapely) reduced mobile energy consumption by 60.4% while maintaining security fidelity in high-risk zones; socio-technical UI friction cut false police escalations by 39.6%.

EXPERIENCE

Cybersecurity AI/ML Researcher - Intern

SwiftSafe

March 2026 – Present

Remote

- Conducting adversarial ML and AI threat research under direct CTO supervision as part of a 6-month structured program.
- Producing VAPT reports and security tooling in Python and Bash targeting AI-integrated systems and threat intelligence pipelines.
- Researching emerging attack surfaces at the intersection of machine learning model security and offensive cybersecurity.

Cyber Security Intern - Ethical Hacking

Anonymous India Cyber Defence Academy

February – March 2026

Remote

- Executed 12 structured penetration testing assessments - network scanning, exploitation, and post-exploitation across virtualized environments.
- Applied SQL Injection, XSS, and Windows Defender evasion using Nmap and Metasploit; authored 12 VAPT reports with custom Bash recon tooling and AI-assisted payloads.
- Awarded Ethical Hacking Internship Certificate for exceptional performance (Credential ID: ANO013698).

Technical Security Writer

InfoSec Writeups · CoinsBench · Block Magnates

2025 – Present

Remote

- Publishing peer-level research articles on blockchain security, smart contract vulnerabilities, and Web3 threat intelligence for a combined readership of 200K+ security professionals.

Founder - Campus Security Chapter

ThunderCipher, MHSSCE

December 2025 – Present

Mumbai

- Founded the college's offensive security chapter; built a 50+ member community for threat intelligence research, CTF training, and bug bounty methodology.

Independent Security Researcher - Bug Bounty

HackerOne · Bugcrowd · Intigriti · Immunefi · Sherlock

June 2020 – Present

Remote

- Disclosed 30+ vulnerabilities (CORS misconfiguration, IDOR, XSS, token approval manipulation) across 15+ web and blockchain platforms.
- Audited 1,000+ lines of Solidity for DeFi protocols (Bima Protocol, Omni Network) - identified re-entrancy flaws, access control gaps, and tx.origin phishing vectors.
- Achieved Rank 2 in Blackbox 01 Web VAPT; Top 10% globally in Snyk Fetch the Flag CTF (154th / 1,539 teams, 1,300 pts).

MHSSCE VAPT Cell - Co-Founder

- Co-founded the college's inaugural VAPT cell, architecting a standard reporting template to streamline vulnerability disclosures for future audits.
- Mentoring junior members by developing and leading training sessions on fundamental offensive security concepts.
- Curating multiple technical offensive security resources to build the cell's internal knowledge base and prepare the team for upcoming assessments.

PROJECTS

RoninScan : On-Chain Threat Intelligence Engine — [Node.js](#) · [Express.js](#) · [Etherscan API](#) · [PDFKit](#) · [Solidity Static Analysis](#)

- Built a full-stack blockchain security platform that auto-retrieves Ethereum smart contract source or bytecode via Etherscan API and performs static vulnerability analysis.
- Scanner engine detects selfdestruct abuse, delegatecall storage hijacking, low-level .call() reentrancy vectors, and tx.origin phishing patterns - assigns Low / Medium / High risk classifications also it generates branded PDF audit reports (PDFKit) with threat-classified findings; tested across Sepolia testnet contracts. Submitted as BE mini-project.

PROTOCOL ZERO - Web3 Security Masterclass — Solidity · EVM Internals · Smart Contract Auditing

- Authored a 10-chapter open-source security curriculum bridging Web2 penetration testing and Web3 smart contract auditing - covering reentrancy (\$10B exploit class), flash loan attacks, on-chain randomness manipulation, and tx.origin identity exploits.
- Structured as a progressive learning franchise: Phase 1 (Paradigm Shift), Phase 2 (Logic Flaws), Phase 3 (Architecture Exploits) - publicly available on GitHub.

NutriPath AI - Bio-Temporal Performance Agent — Python · Gemini LLM · Google Fit · Google Calendar · Async I/O

- Heuristic-first design: local Python decision matrix handles 80% of logic; invokes Gemini LLM only for complex scenarios - achieving <15ms response via async parallel I/O across Google service streams.
- Differential privacy architecture: biometric data (Google Fit) tokenized through a Local Vault pattern - only anonymized stress vectors shared with the LLM, no raw health data exfiltrated.

EDUCATION

B.E. in Computer Science and Engineering - IoT, Cyber Security & Blockchain

Expected May 2027

Anjuman-I-Islam's M.H. Saboo Siddik College of Engineering · University of Mumbai · CGPA: 7.57

CERTIFICATIONS

- **Student SOC Program Foundations** · Microsoft · Jan 2025
- **Foundation Level Threat Intelligence Analyst** · arcX · Oct 2025 · ID: 6320bf63
- **Autopsy Basics & Digital Forensics Basics** · Cybrary · Oct 2025
- **Purple Teaming Fundamentals** · CyberWarFare Labs · May 2024 · ID: 4bac6de7
- **Android Bug Bounty Hunting** · EC-Council · Jun 2025 · ID: 418693
- **Blockchain Basics** · Cyfrin Updraft · Aug 2025 · ID: BBCC-9LGFVVDV3HZXX
- **Build a Secure Google Cloud Network** · Google · May 2024
- **Bounty Hunt 101 — Learn the Art of Breaking** · DevTown × Microsoft Student Chapter · Jan 2026
- **Advanced Persistent Analyst** · KC7: The Cyber Detective Game · Jan 2026 · ID: 40261

AWARDS & RECOGNITION

- Rank 2 — Blackbox 01 Web VAPT: real-world vulnerability identification, exploitation, and documentation.
- Top 10% Global — Snyk Fetch the Flag CTF: 154th / 1,539 teams (DEAD SIGNAL), 1,300 points.
- Top 15% — Hack The Box & TryHackMe: exploit development and reverse engineering.
- Global Pathways AI Hackathon — February 2026: recognized for outstanding AI/ML MVP.
- Smart India Hackathon — 2024 & 2025: qualified internal selection from 500+ teams.

TECHNICAL SKILLS

Offensive Security: Penetration Testing, VAPT, Bug Bounty Hunting, Exploit Development, Reverse Engineering, Purple Teaming, SOC Operations, Incident Response

Blockchain & Web3: Smart Contract Auditing, Solidity, EVM Internals, Re-entrancy / Flash Loan / Access Control Exploits, DeFi Security, Immunefi, Hardhat

AI Security: Adversarial ML, Threat Intelligence, AI Attack Surface Research, LLM Security

Tools: Burp Suite, Metasploit, Nmap, Wireshark, Ghidra, IDA Pro, Autopsy, OWASP ZAP, Remix IDE, Microsoft Sentinel

Programming: Python, JavaScript, Solidity, Bash, Rust (learning), HTML/CSS

Platforms: HackerOne, Bugcrowd, Sherlock, Immunefi, Hack The Box, TryHackMe, Google Cloud, Etherscan API, Ethereum / EVM